

AI Use Case Discovery for Marigold (SRE) Team

Objective

Build AI Products for the SRE team that will directly make operational improvements for the SRE. These improvements can be measured can be seen through different lenses :

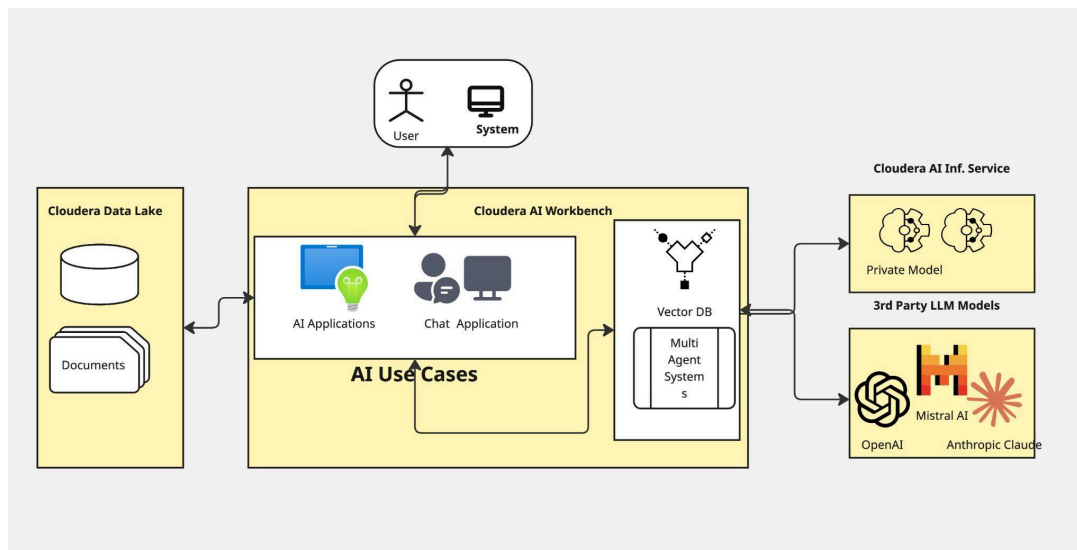
- **Increase the % of automated tasks**
- **Drive improvements in mean response time** by improving quality of information available through Runbooks / SOPs
- **Managing Toil** : Shifting bottlenecks i.e. Improvement in Stability in one area exposes other areas of improvement
- **Root Cause analysis** : Help RCA based on identified issues and then using past history + Code to determine root causes. Which parts of systems to touch

SRE Team Metrics

Overall Metrics that impact SRE Team :

- **Incident Response** : Reducing Mean time to Resolution
- **Manual Toil** : Improve overall % of Automation
- **Deployments** : Lead time to Deploy code artifacts/images etc.
- **Manage per deployment Error Budget** : “how much error / failure can be tolerated around an application/ system before the budget is exceeded and Engineering Team must stabilize application before adding new feature versions

Use Case Implementation Architecture



Potential Use cases to discuss

SRE DOMAIN:

SRE Domain	Use case Description	Business Impact Metric	Components
Incident Management	Autonomous AgenticAI Remediation with Human in Loop: AI Agent uses runbooks and other documents to identify frequently occurring issues and provide “healing” as well as document actions . SRE executes “approval” Example : Typical error (e.g. Package error) involves installation, application restart and redeployment	Improve Mean time to Resolution (decrease MTTR decrease SRE labor cost decrease downtime decrease frequency of escalations and rework)	• AI Inference Service or the Vendor Models • UX based on Gradio/ Flask • Cloudera AI Workbench (AI Agents)
Incident Management	Agent Led Incident Response App : For new, complex, or systemic issues, agents gather context, manage communications, and generate reports, while humans focus on strategic resolution	Improve Mean time to Resolution (decrease incident coordination time decrease cross-team inefficiency decrease major incident duration cost)	• AI Inference Service • UX based on Gradio/ Flask • Cloudera AI Workbench (AI Agents)
Incident Management	Root Cause Analysis with Natural Language Querying : Use combination of RAG system for Runbooks and Agentic System for deeper complex analysis of Agent Logs to provide automated RCA Reports	Lead time (decrease RCA time decrease repeat incidents)	• AI Inference Service • Vector Database • UX based on Gradio/ Flask • Cloudera AI Workbench (AI Agents)
Automation	Build or Optimize Automation Scripts based on Runbooks and Other Documents to create automated scripts that can function	% of Automation in code is improved (leading to improved	• Vector DB • Vibe Coding • Gradio

	as tools for natural language commands such as” Add additional GPU nodes of L40s for horizontally scaling the Payment Processing Cluster”	productivity)	Application • RAG System
Managing Toil	Serf Service Analyzer: Helps provide “self service capabilities” to development teams that are relying on SRE to stabilize their systems. For e.g. answer Natural language commands such as “Provide logs for xx pod for yy applications etc”	% Reduce Manual Toil	• AI Inference Service • UX based on Gradio/ Flask • Cloudera AI Workbench (AI Agents) • Gradio Application
Migration/Upgrade Automation	Automated Migration Agents: Instead of manual job recreation, an agent built in Agent Studio can be programmed to read existing CDE job configurations, adapt them to the newer environment's specs, and trigger the migration. Validation & Verification: A specialized "Validation Agent" can run shadow jobs or smoke tests on the new cluster, compare outputs with the old version, and only "flip the switch" (cut over) once it confirms zero performance regression. This moves Zeta toward the " seamless upgrades " they requested.	Reducing Manual Intervention	• AI Inference Service • UX based on Gradio/ Flask • Cloudera AI Workbench (AI Agents) • Gradio Application
Infrastructure Monitoring	Centralized dashboard within Observability to monitor infrastructure and service health alerts across all environments globally. Proactive alerting for Infra/service downtime. Custom Alerting.	Business Continuity	• AI Inference Service • UX based on Gradio/ Flask • Cloudera AI Workbench (AI Agents) • CM API, Observability

			• Gradio Application
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Questions :

- Please share some high level details of the work done in the team and what are some areas/ challenges from your perspective which should be addressed in this hackathon

Focusing on Value (Impact):

- "Which specific 'Type' of incident accounts for the highest volume of your manual toil today (e.g., resource exhaustion, connectivity, configuration drift)?"
- "If you could reduce the time spent on a single phase of the incident lifecycle (Detection, Identification, or Remediation), which would yield the most ROI?"
- "What is the current 'cost of failure' when an error budget is exceeded? Is it a loss of revenue, or a loss of developer velocity?"

Focusing on Feasibility (Data/Access):

- "Are your Runbooks and SOPs currently stored in a structured format (Markdown/Wiki) or are they tribal knowledge living in Slack and old tickets?"
- "Does the SRE team have existing APIs or CLI tools that an AI Agent can 'hook' into for execution (e.g., Kubernetes API, Terraform, Ansible)?"
- "How 'clean' are your logs? Is there a standardized logging format (JSON) or is it largely unstructured text?"

Datasets

For RAG-based Systems (Knowledge):

- **Documentation:** Export of Confluence/Notion pages, PDF Runbooks, and README files from Git repositories.
- **Historical Incidents:** JSON/CSV exports from PagerDuty, Opsgenie, or ServiceNow or any other Incident Management Systems with "Timeline" and "Resolution" fields.
- **Communication Logs:** Sanitized Slack or Teams threads from past "War Rooms."

For Agentic Analysis (Context/Execution):

- **Telemetry Snapshots:** Prometheus/Grafana snapshots of CPU, Memory, and Network I/O during known outage periods versus "Golden State" periods.

- **Infrastructure-as-Code (IaC):** Terraform or CloudFormation templates (to help the Agent understand the system architecture).
- **Audit Logs:** CloudTrail or K8s Audit logs to help the AI understand "Who changed what" before a failure occurred.

To be Decided

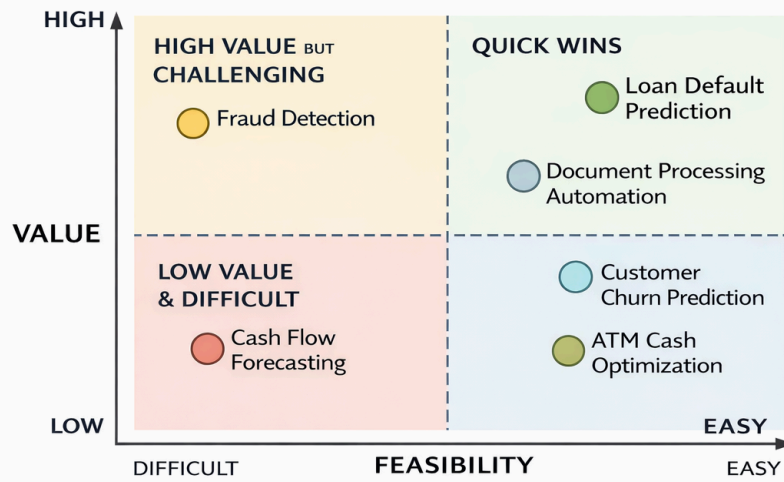
- Where will be the Hackathon System / Environment setup ?
- Confirm the “domain” in which the hackathon use cases will need to be developed
- Hackathon Date

Potential Next Steps:

- Fill the AI Canvas for each of the Selected Use case (See Example Below)

The AI Canvas: Intelligent Invoice Processing		
What task/decision are you examining? Automatically extract key structured attribute data (Invoice Number, Due Date, Supplier) from unstructured or semi-structured documents (PDFs, Scans) across multiple languages and varied formats.		
Prediction Identify the key uncertainty that you would like to resolve. Is the extracted value for a specific attributes (e.g., Invoice Number) correct, and what is the confidence level of this extractions?	Judgment Determine the payoffs to being right versus being wrong. Consider both false positives and false negatives. False Positive (FP): The AI extracts the wrong value but assigns High Confidence. Payoff: Incorrect data enters ERP, requiring costly manual reconciliation/correction later. False Negative (FN): The AI extracts the correct value but assigns Low Confidence. Payoff: Document is flagged for manual review (HITL), wasting staff time but ensuring data quality. Priority: Minimize FPs, as FNs are caught by the HITL process.	Action What are the actions that can be chosen? • Invoice Metadata with high confidence of extraction above a threshold should have Straight through Processing o ERP. • Invoice Data with low confidence should be in an Exception Management Queue for Human Processing • Human Processed Invoices should be used for Retraining
Outcome Choose the measure of performance that you want to use to judge whether you are achieving your outcomes. 1. F1 Score/Accuracy: Must be >95% on production data. 2. Straight-Through Processing (STP) Rate: Percentage of documents processed without human intervention. 3. Time-to-Value: Average time from document ingestion to data available for retrieval.	Training What data do you need on past inputs, actions and outcomes in order to train your AI and generate better predictions? Historical PDFs/Scans, along with the corresponding correct attribute values (labeled data). The key is the user-corrected data from the HITL process.	Input What data do you need to generate predictions once you have an AI algorithm trained? Raw PDF or image file (Ingested via Email, or UI drop).
Feedback How can you use measured outcomes along with input data to generate improvements to your predictive algorithm? The manual corrections made by AP staff in the HITL Exception UI are captured, labeled, and used to continuously retrain and fine-tune the AI model on new or problematic document formats.	How will this AI impact on the overall workflow? Explain here how the AI for this task/decision will impact on related tasks in the overall workflow Will it involve staff retraining or job redesign? The role of the AP/Sales staff shifts from Data Entry to Exception Handling, Validation, and Model Training. This requires job redesign and retraining for efficiency gains.	

- Prioritize among Use Cases between Business Value (Why?) and Technical Feasiblity (How) (see example below)



- IN BANKING DOMAIN
- | | |
|---|---|
|  Fraud Detection |  ATM Cash Optimization |
|  Cash Flow Forecasting |  Customer Churn Preds |